

BALA VENKATARAJU SINGAMPALLI

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SUMMARY

Entry-level Software Engineer with a strong foundation in full-stack development (MERN) and backend engineering. Experienced building and deploying production-style systems — microservice architectures, computer-vision pipelines, and speech-processing systems — using React, Node.js, Express.js, and Django. Proficient in REST API design, SQL/NoSQL databases, containerization, and cloud deployment (Vercel, Railway, Render).

TECHNICAL SKILLS

- Frontend: HTML, CSS, Tailwind CSS, React.js, JavaScript (Vanilla JS)
- Backend: Node.js, Express.js, Django, FastAPI, REST APIs, WebSockets
- Databases: MySQL, MongoDB, Redis
- Languages: JavaScript, Python
- Tools & DevOps: Git, GitHub, Docker, Nginx, Postman, ffmpeg
- Cloud & Deployment: Vercel, Railway, Render, MongoDB Atlas, Supabase
- AI/ML (applied): YOLOv8, InsightFace, OpenAI Whisper, OpenCV, PyTorch

PROJECTS

EduVision — AI Face Recognition Attendance System | **live**

Python, FastAPI, InsightFace, Node.js, Express, React, MySQL, MongoDB, Redis, Cloudinary, Nginx, Docker

- Built a full-stack attendance system using 5 independent microservices (Auth, Attendance, Report, WhatsApp, Face Recognition) behind an Nginx API gateway, reducing per-class attendance time from 10 minutes to 30 seconds (95% faster)
- Implemented AI-based face recognition using InsightFace (ArcFace) with cosine similarity matching, achieving 99.2% model accuracy, with manual verification for low-confidence matches
- Designed a polyglot persistence architecture (MySQL, MongoDB, Redis, Cloudinary), JWT authentication, and cloud deployment (Vercel + Railway) load-tested for 500+ concurrent users

Speech Recognition Web Application — Fine-Tuned Whisper

Python, Django, PyTorch, OpenAI Whisper, ffmpeg, pytsx3

- Engineered a full-stack speech transcription system around a Whisper model fine-tuned on a custom dataset of thousands of labeled .wav audio samples
- Built an audio processing pipeline (ffmpeg-based decoding, file upload, real-time microphone recording) feeding transcription and text-to-speech (TTS) output via Django routes
- Implemented temp-file cleanup logic and structured configuration management for a 900MB fine-tuned model

INTERNSHIP EXPERIENCE

Vizag Steel Plant (Rashtriya Ispat Nigam Ltd.) — Software/AI Intern

Jun 2024 – Jul 2024

Python, Django, YOLOv8, OpenCV, Tailwind CSS | Live demo deployed on Render

- Built and deployed a full-stack PPE detection web app supporting real-time webcam detection, image upload, and a processed-image gallery
- Integrated a YOLOv8 object detection model to identify helmets, vests, and gloves across uploaded images and live camera feeds
- Designed a responsive Django + Tailwind CSS interface with dark mode, client/server-side validation, and Django ORM-backed image gallery

EDUCATION

Acharya Nagarjuna University, Guntur

2022 – 2026

B.Tech in Artificial Intelligence & Machine Learning — CGPA: 8.2/10

CODING PROFILES

- GeeksforGeeks: [balavenkataraj75f4](#)
- LeetCode: [Rajudotin](#)